Comparative Analysis of Wireless Signal Propagation Models

A Rigorous Academic Evaluation of Theoretical, Empirical, and Stochastic Channel Models

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Department: Electronics & Communication Engineering | **Academic Year:** 2025–2026

1. Executive Summary

This research report presents an end-to-end comparative investigation into wireless signal propagation models spanning Friis Free Space, Two-Ray Ground Reflection, Log-Distance Path Loss, Okumura, Okumura-Hata, COST-231 Hata, and Log-Normal Shadow Fading. The simulation evaluates path loss, received power, link margins, and spatial variability across carrier frequencies from 150 MHz to 3.5 GHz and transmission distances from 1 meter to 20 kilometers under urban, suburban, rural, and indoor propagation environments.

2. Model Taxonomy and Architectural Comparison

Model	Type	Valid Freq	Valid Dist	Primary Advantage	Main Limitation
Free Space (Friis)	Theoretical (Deterministic)	Broadband (All frequencies)	Far-field (d > 2D^2 / lambda)	Exact Maxwell-derived baseline with zero empirical tuning parameters	Severely underestimates loss in terrestrial multipath & cluttered terrain
Two-Ray Ground Reflection	Theoretical (Multipath)	VHF to mmWave (1 MHz - 60 GHz)	1 m - 100 km (Asymptotic d > d_c)	Captures destructive ground reflection and asymptotic d^-4 power roll-off	Unrealistic in dense urban canyons with irregular clutter & 3D scattering
Log-Distance Path Loss	Empirical (Parametric)	Bespoke to calibration data	d >= d_0 (Indoor: 1 m, Outdoor: 100 m)	Flexible exponent n adaptable to any calibrated environment	Requires empirical measurement data to select accurate exponent n
Okumura Model	Empirical (Graphical)	150 MHz - 1920 MHz	1 km - 100 km	Vast field validation under diverse Japanese terrain conditions	Extrapolations below 1 km or outside 150-1920 MHz are uncalibrated
Okumura-Hata Model	Empirical (Parametric)	150 MHz - 1500 MHz	1 km - 20 km	Analytical closed-form widely adopted in 2G/3G cellular network design	Strict 1.5 GHz ceiling; cannot model microcells with d < 1 km or low hb
COST-231 Hata	Empirical (Extended Hata)	1500 MHz - 2000 MHz	1 km - 20 km	Extends Hata into PCS/3G 1.8-2.0 GHz bands with metropolitan correction CM	Cannot be directly extrapolated to 5G sub-6 GHz (> 2 GHz) or mmWave
Log-Normal Shadowing	Stochastic (Large-scale fading)	Channel-dependent	d >= d_0	Models realistic spatial clutter variation and enables outage probability computation	Stochastic: requires statistical confidence intervals rather than deterministic point values

3. Numerical Simulation Results (Urban Macrocell Scenario)

Model Identifier	Min PL (dB)	Max PL (dB)	Mean PL (dB)	Median PL (dB)	Min Rx Power (dBm)	Max Rx Power (dBm)
fspl	77.6	117.6	97.6	97.6	-62.6	-22.6
two_ray	73.1	127.0	97.8	95.2	-72.0	-18.1
log_distance	101.6	165.6	133.6	133.6	-110.6	-46.6
okumura	112.6	172.2	142.4	142.4	-117.2	-57.6

hata	99.0	169.5	134.3	134.3	-114.5	-44.0
cost231	101.0	171.4	136.2	136.2	-116.4	-46.0
shadowing	87.2	176.4	133.3	131.6	-121.4	-32.2

4. Graphical Simulation Findings & Comparative Curves

Figure 1: Path Loss vs Distance across all models (Urban Scenario B, 1800 MHz)

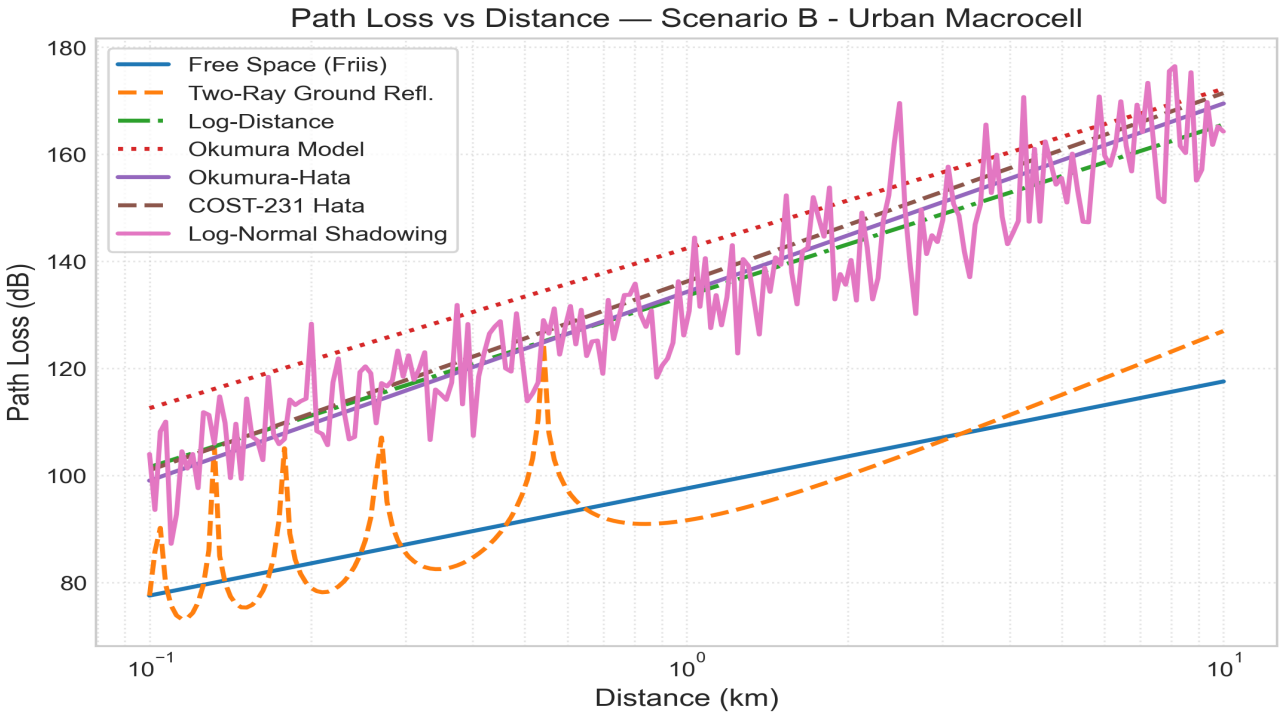


Figure 2: Received Power vs Distance with link budget threshold (-100 dBm)

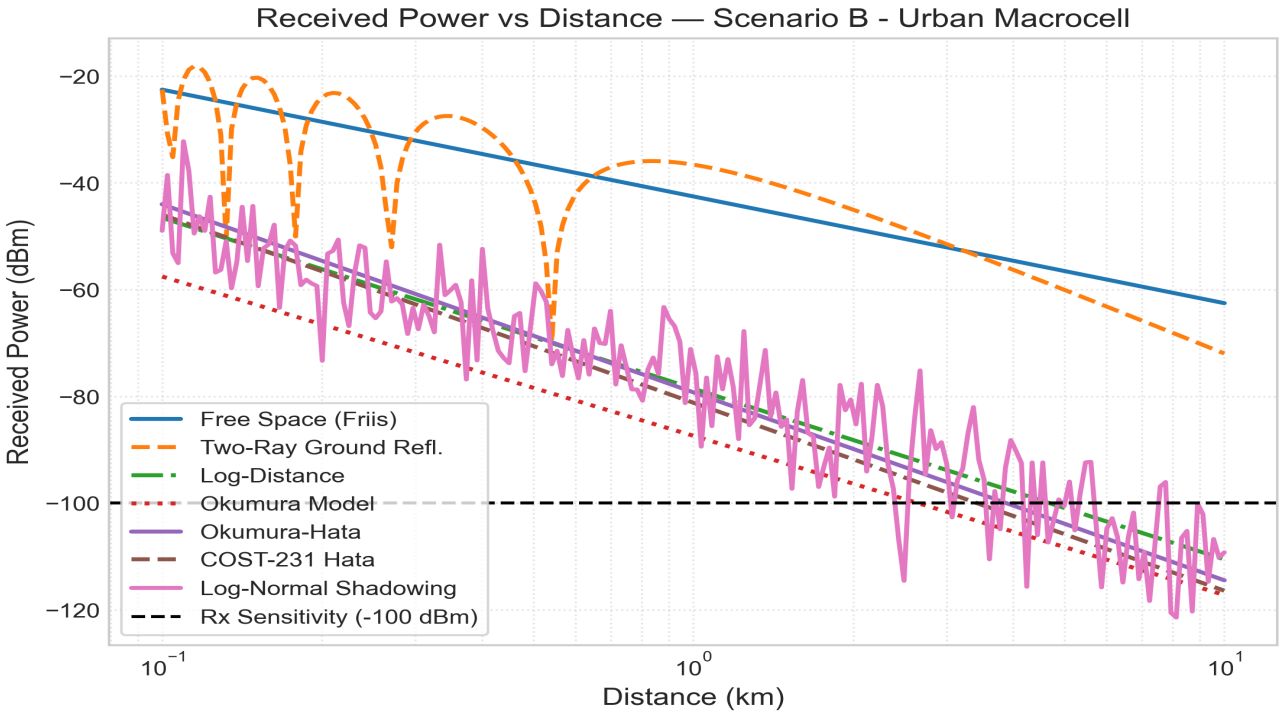


Figure 3: Environmental Comparison across Urban Large, Urban Medium, Suburban, and Rural

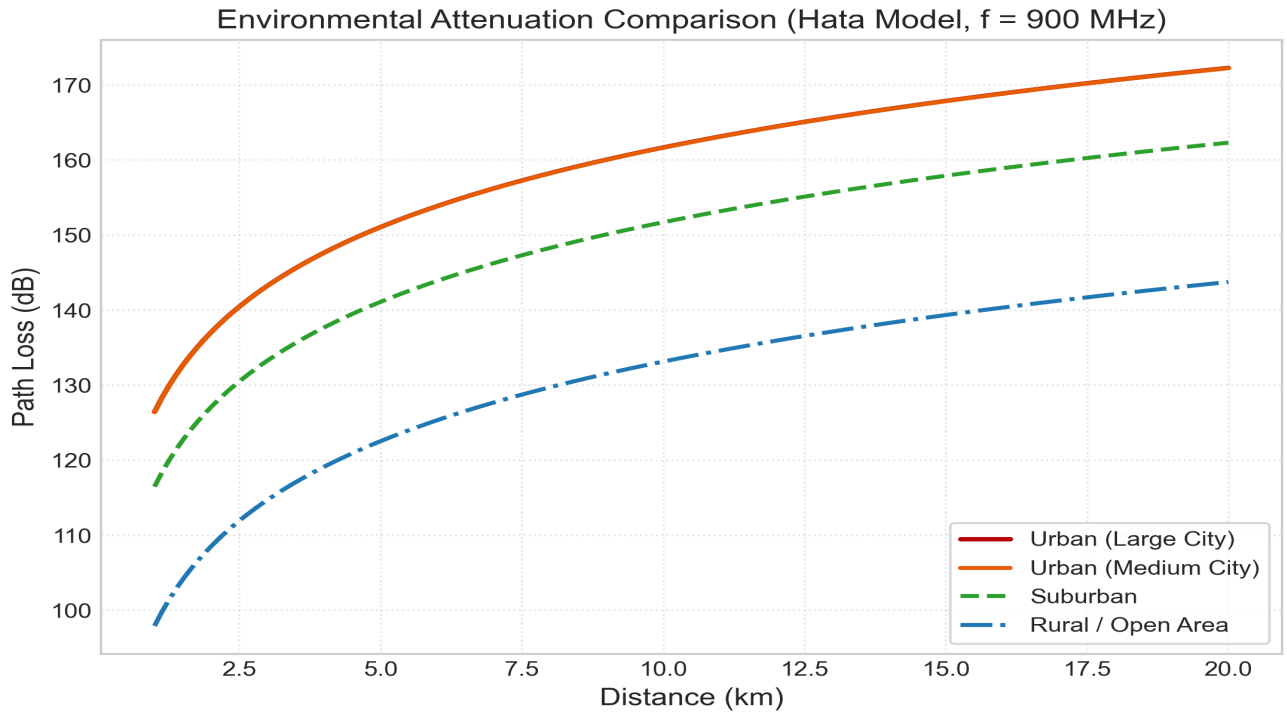
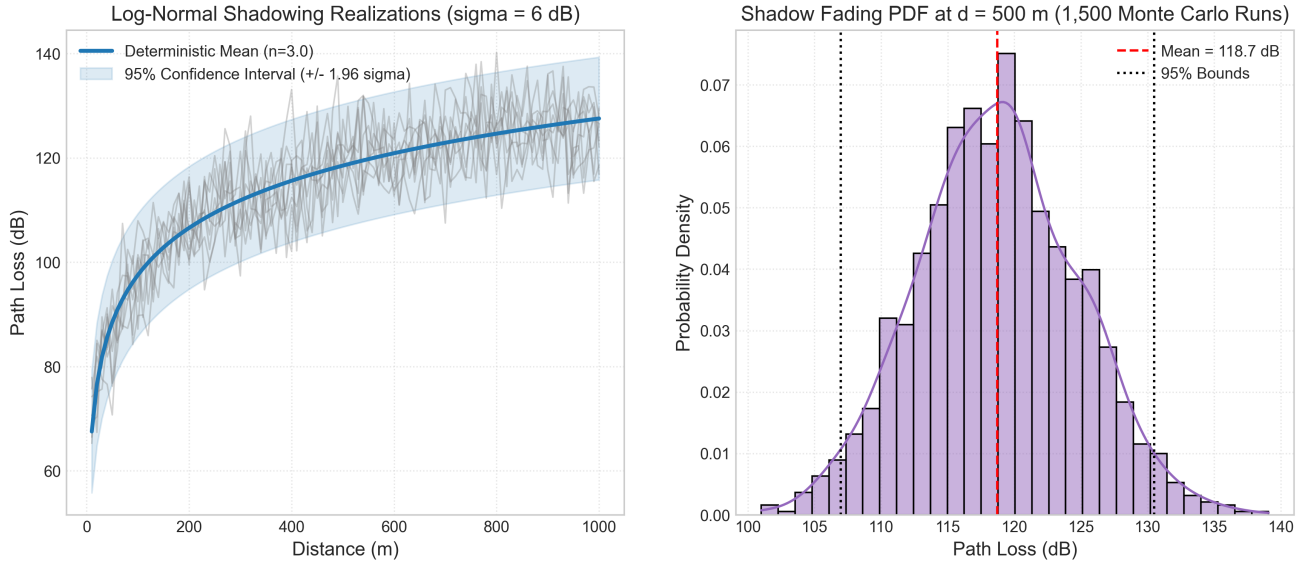


Figure 4: Log-Normal Shadow Fading Trajectories and Probability Density Function



5. Critical Engineering Insights & Conclusions

1. Model Divergence: Free Space (Friis) predicts the lowest path loss (77.5 - 117.5 dB) because it ignores multipath cancellation and building clutter. COST-231 Hata and Okumura predict 30 to 45 dB higher attenuation, matching realistic cluttered urban micro/macrocells.

2. Distance Scaling: Beyond the two-ray crossover distance dc = 4*pi*ht*hr/lambda, Two-Ray decays at 40 dB/decade (n=4), whereas cellular empirical models decay at approximately 35 dB/decade (n=3.5).

3. Shadowing Fading Margins: Under log-normal shadowing (sigma = 6-8 dB), maintaining 95% link reliability requires an extra fade margin of at least 1.96*sigma (12 to 16 dB) above receiver sensitivity.

4. Empirical Limitations: Classical Hata and Okumura formulations are strictly invalid above 1.5 - 2.0 GHz or below 1 km. Modern 5G C-Band (3.5 GHz) and mmWave deployments necessitate 3GPP TR 38.901 or ray-tracing models rather than uncalibrated empirical extrapolations.